Göktuğ Arda Gök

Istanbul, Turkey | goktug@admex.com.tr | goktug-adm.github.io/portfolio

EDUCATION

Sabancı University, B.Sc. in Mechatronics Engineering — Sep 2021 – Jan 2026
Istanbul, Turkey. Taught in English, 240 ECTS.
Relevant coursework: Control System Design, Systems Modelling and Control, Industrial Control, Mechatronics System Design, Autonomous Mobile Robotics, Electronic Circuits I and II, Dynamics, Mechanical Systems, Manufacturing Processes, Machine Learning, Artificial Intelligence.

Lycée Saint-Joseph, High School Diploma, Istanbul — 2016 – 2021

WORK EXPERIENCE

Software and Mechatronics Engineer, ADMEX Group (Swedish branch) — Jan 2026 – Present
Istanbul, Turkey. Industrial automation company building robotic positioners, tracks and gantries.

- Worked on gantry drive and control systems: servo motor and cycloidal gearbox selection, encoder specifications, control-panel architecture (teach pendant, motion card, fieldbus modules), I/O planning and explosion-proof (Ex) motor variants with SEW-Eurodrive.
- Part and assembly modelling in Siemens NX for positioner and gantry components with the engineering team.
- Developing a parametric product configurator where customers enter product dimensions, preview the product in 3D and download CAD models, based on STEP exports from Siemens NX and Python geometry processing.
- Built ADMEX Web, a pricing and configuration platform for the sales team with PDF quote generation and AI-supported sales analysis (Next.js, TypeScript, MongoDB).
- Automated bulk edits across the 468-page product catalog with Python.

Engineering Intern, Altınay Defence — Jun 2025 – Aug 2025

- Built a swarm drone simulation in MATLAB App Designer (formation control, obstacle scenarios) and in ROS 2 (custom coordination nodes, RViz visualisation).

Intern, FANUC Turkey — Jul 2024 – Aug 2024

- Trained in industrial robot programming, CNC maintenance and repair, and Roboguide simulation (Level 1).

Intern, ADMEX Advanced Motion Experts — Jul 2023

PATENTS AND DESIGN REGISTRATIONS

- Full Protection Robot Slider System. TR 2024/010626 and TR 2024/010627 (Dec 2024), international filing PCT/TR2024/051751. Inventor and applicant.
- Robotic Linear Motion Unit. TR 2022/014378 (Nov 2022). Inventor and applicant.

Verification: TÜRKPATENT record

SELECTED PROJECTS

- Graduation project: analysis, build and real-time control of a parallel spherical manipulator (Simulink/QUARC over UDP), with a vision-based orientation tracker on a Raspberry Pi 5 using a global-shutter camera, ArUco markers and the Kabsch algorithm (30 Hz, sub-degree accuracy).
- Quadrotor dynamics for 4-DOF and 6-DOF systems and an ultrasonic range-finder control system in MATLAB/Simulink.
- Beckhoff motor setup programmed in Structured Text, and a PLC-controlled painting and drying line.
- Fully functional V12 engine assembly in SolidWorks, built in a team of four (crankshaft, pistons, connecting rods, valve train).
- 3D-printed 1:10 positioner and track models for the Schweissen & Schneiden trade fair.

SKILLS

Engineering: SolidWorks (advanced), Siemens NX, MATLAB/Simulink, ROS 2, Roboguide, PLC (ST)
Software: AI-assisted development with Claude Code, Cursor and ChatGPT, used to build web tools (Next.js, TypeScript, MongoDB) and Python automation scripts. Basic Python and C++, Git, Linux.
Languages: Turkish (native), English (C1, IELTS Academic 7.5), French (B2, DELF)

ACTIVITIES

- Captain, Sabancı Seahawks Football Team (2021–2025). Board Member, University Sports Club (2024–2025).
- Volunteer, Civic Involvement Project (2022). Debate Club, Lycée Saint-Joseph (2018).